The Gaussian Correlation Inequality after Royen

after T. Royen (2014) and R. Latała & D. Matlak (2015)

1 Introduction

Theorem 1.1 (Gaussian correlation inequality)**.** For any closed symmetric convex sets $K, L \subseteq \mathbb{R}^d$ and any centered Gaussian measure μ on $\mathbb{R}^d$,

$$\mu(K \cap L) \;\geqslant\; \mu(K)\,\mu(L). \tag{1}$$

These notes prove Theorem 1.1 by Royen's analytic argument, following Latała–Matlak's Gaussian-only exposition. The inputs are the Laplace transform of a squared Gaussian, a determinant expansion, dominated convergence, and integration by parts. Historical background and further remarks are collected in Section 6.

Notation 1.2.
- $\mathcal{N}(0, C)$: centered Gaussian distribution with covariance C.
- $M_{n\times m}$: real $n \times m$ matrices.
- For $A = (a_{ij})_{i,j\leqslant n}$ and $J \subseteq [n] := \{1, \dots, n\}$, $A_J := (a_{ij})_{i,j\in J}$ is the principal submatrix, $|A|$ the determinant, $|J|$ the cardinality.
- I_n: $n \times n$ identity.
- For $\varepsilon \in \{-1, 1\}^n$, $x \in (0, \infty)^n$: $\sqrt{x}_\varepsilon := (\varepsilon_i\sqrt{x_i})_{i\leqslant n}$.

2 Setup and reduction to rectangles

Definition 2.1 (Centered Gaussian measure)**.** For $C \succ 0$, the measure $\mathcal{N}(0, C)$ on $\mathbb{R}^n$ has density

$$\phi_C(x) \;=\; (2\pi)^{-n/2}|C|^{-1/2}\exp\bigl(-\tfrac{1}{2}\langle C^{-1}x, x\rangle\bigr).$$

We assume $C \succ 0$ throughout the proof; the extension to singular $C \succeq 0$ is deferred to Section 6.2.

Definition 2.2. $K \subseteq \mathbb{R}^n$ is *symmetric* if $K = -K$. A *symmetric slab* is $S_{v,t} := \{x \in \mathbb{R}^n : |\langle v, x\rangle| \leqslant t\}$ for $v \in \mathbb{R}^n \setminus \{0\}$, $t > 0$.

Lemma 2.3 (Slab approximation)**.** Every closed symmetric convex set $K \subseteq \mathbb{R}^d$ is a countable intersection of symmetric slabs $S_{v,t}$.

Proof. Step 1: reduction to bounded K. Set $K_m := K \cap [-m, m]^d$, a closed symmetric convex set. Since $K = \bigcup_m K_m$, by monotone convergence $\mu(K) = \lim_m \mu(K_m)$ and similarly for $K \cap L$, so it suffices to prove the slab approximation for bounded K; we assume $K \subseteq B(0, R)$ for some $R < \infty$ below.

Step 2: bounded case. Let $\mathcal{V} := \{(v, t) \in \mathbb{Q}^d \times \mathbb{Q}_{>0} : K \subseteq S_{v,t}\}$, countable. Trivially $K \subseteq \bigcap_{(v,t)\in\mathcal{V}} S_{v,t}$.

Conversely, fix $y \notin K$ and let $\varepsilon > 0$ with $B(y, \varepsilon) \cap K = \varnothing$. Strict separation of two disjoint closed convex sets (one compact) yields $w \in \mathbb{R}^d \setminus \{0\}$ and $a \in \mathbb{R}$ with $\sup_{x\in K}\langle w, x\rangle \leqslant a$ and $\langle w, y\rangle \geqslant a + \varepsilon|w|$.

Since $0 \in K$ we have $a \geqslant 0$; and $K = -K$ gives $\sup_{x\in K} |\langle w, x\rangle| \leqslant a$. Pick $(w', a') \in \mathbb{Q}^d \times \mathbb{Q}_{>0}$ with $|w' - w| < \delta$, $|a' - a| < \delta$. Then

$$\sup_{x\in K} |\langle w', x\rangle| \leqslant a + \delta R, \qquad |\langle w', y\rangle| \geqslant a + \varepsilon|w| - \delta|y|.$$

Choosing $\delta < \min(\varepsilon|w|/(|y| + R + 1), \varepsilon|w|/2)$ and $a' := a + \delta R + \delta/2 \in \mathbb{Q}_{>0}$ gives $K \subseteq S_{w',a'}$ and $y \notin S_{w',a'}$. □

By Theorem 2.3 and monotone convergence, the GCI reduces to the case where K and L are finite intersections of symmetric slabs. Writing

$$K = \big\{x : |\langle v_i, x\rangle| \leqslant t_i,\ 1 \leqslant i \leqslant n_1\big\}, \qquad L = \big\{x : |\langle v_i, x\rangle| \leqslant t_i,\ n_1{+}1 \leqslant i \leqslant n\big\},$$

setting $n = n_1 + n_2$ and $X_i := \langle v_i, G\rangle$ for $G \sim \mu$, the GCI reduces to:

Theorem 2.4 (GCI, analytic form)**.** Let $X = (X_1, \dots, X_n)$ be centered Gaussian with $C \succ 0$, $n = n_1 + n_2$. For any $t_1, \dots, t_n > 0$,

$$\mathsf{P}(|X_i| \leqslant t_i,\ i \leqslant n) \ \geqslant\ \mathsf{P}(|X_i| \leqslant t_i,\ i \leqslant n_1)\,\mathsf{P}(|X_i| \leqslant t_i,\ n_1{<}i \leqslant n). \tag{2}$$

3 Auxiliary lemmas

3.1 Laplace transform of squared Gaussian vectors

Lemma 3.1. For $X \sim \mathcal{N}(0, C)$ in $\mathbb{R}^n$ and $\lambda_i \geqslant 0$,

$$\mathsf{E}\exp\Big(-\sum_{i=1}^{n} \lambda_i X_i^2\Big) \ =\ |I_n + 2\Lambda C|^{-1/2}, \qquad \Lambda := \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Proof. The Gaussian integral $\int_{\mathbb{R}^n} e^{-\langle Ax,x\rangle}\,\mathrm{d}x = \pi^{n/2}|A|^{-1/2}$ (for $A \succ 0$) gives, for $Y \sim \mathcal{N}(0, I_n)$ and any symmetric B with $I_n - 2B \succ 0$,

$$\mathsf{E}e^{\langle BY,Y\rangle} = (2\pi)^{-n/2} \int e^{-\langle(\frac{1}{2}I_n - B)x,x\rangle}\,\mathrm{d}x = |I_n - 2B|^{-1/2}. \tag{3}$$

Since $C \succ 0$, there exists $A \in M_{n\times n}$ with $C = AA^\top$ (e.g. $A = C^{1/2}$); then $X := AY \sim \mathcal{N}(0, C)$. Apply (3) with $B := -A^\top\Lambda A$. For $\lambda_i \geqslant 0$, $B \preceq 0$ so $I_n - 2B \succeq I_n \succ 0$ and (3) applies. Then

$$\mathsf{E}\exp\Big(-\sum_i \lambda_i X_i^2\Big) = \mathsf{E}\exp(-\langle A^\top\Lambda A\,Y, Y\rangle) = |I_n + 2A^\top\Lambda A|^{-1/2}.$$

Finally, $|I_n + A_1A_2| = |I_n + A_2A_1|$ (Sylvester's determinant identity) with $A_1 := A^\top$, $A_2 := 2\Lambda A$ yields $|I_n + 2A^\top\Lambda A| = |I_n + 2\Lambda AA^\top| = |I_n + 2\Lambda C|$. □

3.2 A determinant identity

Lemma 3.2. (i) For $A \in M_{n\times n}$,

$$|I_n + A| \ =\ 1 + \sum_{\varnothing\neq J\subseteq[n]} |A_J|. \tag{4}$$

(ii) For $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22}\end{smallmatrix}\right)$ with A_{ii} invertible,

$$|A| = |A_{11}||A_{22}| \cdot \big|I_{n_1} - A_{11}^{-1/2}A_{12}A_{22}^{-1}A_{21}A_{11}^{-1/2}\big|. \tag{5}$$

If $A \succ 0$, then $0 \leqslant A_{11}^{-1/2}A_{12}A_{22}^{-1}A_{21}A_{11}^{-1/2} \leqslant I_{n_1}$.

Proof. (i). By Leibniz, $|I_n + A| = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + a_{i\sigma(i)})$. Expanding the product, each summand corresponds to a choice of $J \subseteq [n]$ (positions where $a_{i\sigma(i)}$ is selected); non-zero contributions require $\sigma(i) = i$ for $i \notin J$. The σ's fixing $[n] \setminus J$ correspond to permutations of J, giving $\sum_J |A_J|$; the $J = \varnothing$ term is 1.

(ii). Write $A_{11}^{1/2}, A_{22}^{1/2}$ for the positive symmetric square roots. With $P := A_{11}^{-1/2} A_{12} A_{22}^{-1/2}$ and $Q := A_{22}^{-1/2} A_{21} A_{11}^{-1/2}$,

$$A = \operatorname{diag}(A_{11}^{1/2}, A_{22}^{1/2}) \left(\begin{smallmatrix} I & P \\ Q & I \end{smallmatrix}\right) \operatorname{diag}(A_{11}^{1/2}, A_{22}^{1/2}).$$

Row-reducing the middle block matrix by $R_1 \leftarrow R_1 - PR_2$ gives $\left(\begin{smallmatrix} I-PQ & 0 \\ Q & I \end{smallmatrix}\right)$ with determinant $|I - PQ|$; taking determinants yields (5).

When $A \succ 0$: for all $s, t \in \mathbb{R}$ and $x \in \mathbb{R}^{n_1}$, $y \in \mathbb{R}^{n_2}$,

$$\langle A(sx, ty)^\top, (sx, ty)^\top \rangle = s^2 \langle A_{11}x, x \rangle + 2st \langle A_{21}x, y \rangle + t^2 \langle A_{22}y, y \rangle \geqslant 0.$$

Nonnegativity of this quadratic in (s, t) forces $\langle A_{21}x, y \rangle^2 \leqslant \langle A_{11}x, x \rangle \langle A_{22}y, y \rangle$. Setting $x := A_{11}^{-1/2}u$, $y := A_{22}^{-1/2}v$: $\langle Qu, v \rangle^2 \leqslant |u|^2 |v|^2$, i.e., $\|Q\|_{\mathrm{op}} \leqslant 1$, equivalently $0 \preceq Q^\top Q \preceq I_{n_1}$. For A symmetric, $Q = P^\top$, so $A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} = PP^\top = Q^\top Q$. □

3.3 Density of $Z(\tau)$ and the interchange lemma

Partition

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad C_{ij} \in M_{n_i \times n_j},\ n_1 + n_2 = n, \tag{6}$$

and define the interpolating family

$$C(\tau) := \begin{pmatrix} C_{11} & \tau C_{12} \\ \tau C_{21} & C_{22} \end{pmatrix} = \tau C(1) + (1 - \tau) C(0), \qquad \tau \in [0, 1]. \tag{7}$$

Since $C(0) = \operatorname{diag}(C_{11}, C_{22}) \succ 0$ and $C(1) = C \succ 0$, each $C(\tau) \succ 0$ by convexity. Let $X(\tau) \sim \mathcal{N}(0, C(\tau))$ and $Z_i(\tau) := \frac{1}{2} X_i(\tau)^2$.

Lemma 3.3 (Density and interchange)**.** Let $f(x, \tau)$ denote the density of $Z(\tau)$ on $(0, \infty)^n$.

(i)

$$f(x, \tau) = |C(\tau)|^{-1/2} (4\pi)^{-n/2} \frac{1}{\sqrt{x_1 \cdots x_n}} \sum_{\varepsilon \in \{-1,1\}^n} e^{-\langle C(\tau)^{-1} \sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon \rangle} \mathbf{1}_{(0,\infty)^n}(x). \tag{8}$$

(ii) For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geqslant 0$,

$$\int_K e^{-\sum_i \lambda_i x_i} \frac{\partial}{\partial \tau} f(x, \tau) \, \mathrm{d}x \;=\; \frac{\partial}{\partial \tau} \int_K e^{-\sum_i \lambda_i x_i} f(x, \tau) \, \mathrm{d}x. \tag{9}$$

Proof. (i). The density of $X(\tau)$ is $\phi_{C(\tau)}$ from Theorem 2.1. The map $y \mapsto x = (\frac{1}{2} y_i^2)_i$ is a bijection $(\mathbb{R} \setminus \{0\})^n \to (0, \infty)^n$ on each of the 2^n sign orthants, with Jacobian $\prod_i |y_i|^{-1} = \prod_i (2x_i)^{-1/2}$. Summing over sign patterns $\varepsilon \in \{-1, 1\}^n$ (where $y_i = \varepsilon_i \sqrt{2x_i}$) and using $\langle C(\tau)^{-1} y, y \rangle = 2 \langle C(\tau)^{-1} \sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon \rangle$ gives (8).

(ii). The family $\{C(\tau) : \tau \in [0, 1]\}$ is a compact subset of $\{C' \succ 0\}$, so there is $a_0 > 0$ with $C(\tau) \succeq a_0 I_n$ for all τ, equivalently $C(\tau)^{-1} \succeq a_0' I_n$ for some $a_0' > 0$ (the smallest eigenvalue of $C(\tau)^{-1}$ is continuous in τ). Since $\langle C(\tau)^{-1} \sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon \rangle \geqslant a_0' |\sqrt{x}_\varepsilon|^2 = a_0' \sum_i x_i$, set $a := a_0'$. Similarly, $\tau \mapsto |C(\tau)|^{-1/2}$ and $\tau \mapsto C(\tau)^{-1}$ are C^1 on $[0, 1]$, so their τ-derivatives are bounded in operator norm, giving $M < \infty$ and $b < \infty$ with the claimed bounds (the derivative formula $\frac{\mathrm{d}}{\mathrm{d}\tau} C(\tau)^{-1} = -C(\tau)^{-1} C'(\tau) C(\tau)^{-1}$ follows from differentiating $C(\tau) C(\tau)^{-1} = I_n$, and $C'(\tau) = C(1) - C(0)$ is constant).

Differentiating (8) term by term,

$$\left|\partial_\tau f(x,\tau)\right| \leqslant c_1\big(M + b\textstyle\sum_i x_i\big)\frac{1}{\sqrt{x_1\cdots x_n}}e^{-a\sum_i x_i}\cdot 2^n \leqslant g(x) := c\,\frac{1+\sum_i x_i}{\sqrt{x_1\cdots x_n}}e^{-a\sum_i x_i},$$

uniformly in $\tau \in [0,1]$, with c depending only on C, n. Since $\int_0^\infty x^{-1/2}(1+x)e^{-ax}\,\mathrm{d}x < \infty$ and the n-dimensional integrand factors, $\int_{(0,\infty)^n} g(x)\,\mathrm{d}x < \infty$. As $e^{-\sum_i \lambda_i x_i} \leqslant 1$, the bound $e^{-\sum_i \lambda_i x_i}|\partial_\tau f| \leqslant g$ persists, and dominated convergence justifies interchanging $\frac{\mathrm{d}}{\mathrm{d}\tau}$ and $\int_K$ to give (9). □

3.4 The noncentral gamma and the h-family

Definition 3.4. For $\alpha > 0$, $y \geqslant 0$, the *noncentral gamma density* is

$$g_\alpha(x,y) := e^{-x-y}\sum_{k=0}^{\infty}\frac{x^{k+\alpha-1}}{\Gamma(k+\alpha)}\,\frac{y^k}{k!}, \qquad x > 0.$$

Remark 3.5. Equivalently,

$$g_\alpha(x,y) = \sum_{k\geqslant 0} e^{-y}\tfrac{y^k}{k!}\,g_{\alpha+k}(x),$$

a Poisson(y) mixture of central gamma($\alpha+k$) densities; so $g_\alpha(\cdot, y)$ is a probability density for each $y \geqslant 0$.

Definition 3.6. For $\mu > 0$, $\alpha \in (0,\infty)^n$, and $Y = (Y_1,\dots,Y_n)$ with $Y_i \geqslant 0$ a.s.,

$$h_{\alpha,\mu,Y}(x) := \mathsf{E}\prod_{i=1}^{n}\tfrac{1}{\mu}\,g_{\alpha_i}\big(\tfrac{x_i}{\mu}, Y_i\big), \qquad x \in (0,\infty)^n.$$

Lemma 3.7 (Properties of the h-family)**.** Write $h_\alpha := h_{\alpha,\mu,Y}$.

(i) $h_\alpha \geqslant 0$ and $\int h_\alpha\,\mathrm{d}x = 1$.

(ii) If $\alpha_i > 1$, then $h_\alpha(x) \to 0$ as $x_i \to 0+$, and

$$\frac{\partial}{\partial x_i}h_\alpha = h_{\alpha-e_i} - h_\alpha. \tag{10}$$

(iii) If $\alpha \in (1,\infty)^n$, then for any $J \subseteq [n]$ and $\lambda_i \geqslant 0$, $\partial^{|J|}h_\alpha/\partial x_J \in L_1((0,\infty)^n)$ and

$$\int e^{-\sum_i \lambda_i x_i}\frac{\partial^{|J|}}{\partial x_J}h_\alpha\,\mathrm{d}x \;=\; \Big(\prod_{i\in J}\lambda_i\Big)\int e^{-\sum_i \lambda_i x_i}h_\alpha\,\mathrm{d}x. \tag{11}$$

Proof. (i). $\int_0^\infty \mu^{-1}g_\alpha(x_i/\mu, Y_i)\,\mathrm{d}x_i = \int_0^\infty g_\alpha(u, Y_i)\,\mathrm{d}u = 1$ for each ω (Theorem 3.5); Fubini gives $\int h_\alpha = \mathsf{E}[1] = 1$.

(ii). Gamma bound. Γ is strictly convex on $(0,\infty)$ (its second derivative on log-scale is positive) with unique minimum at some $s_0 \in (1,2)$ and $\Gamma(s_0) > \frac{1}{2}$ (numerically $s_0 \approx 1.462$, $\Gamma(s_0) \approx 0.886$), so

$$\Gamma(s) \geqslant \tfrac{1}{2} \quad \text{for all } s > 0. \tag{12}$$

For $k = 0$, (12) gives $\Gamma(\alpha)^{-1} \leqslant 2$. For $k \geqslant 1$, $\alpha > 0$, induction on k via $\Gamma(k+\alpha) = (k-1+\alpha)\Gamma(k-1+\alpha)$ yields $\Gamma(k+\alpha) \geqslant \frac{1}{2}(k-1)!$. Substituting into $g_\alpha(x,y)$ and using $y^k e^{-y}/k! \leqslant 1$, $\sum_{k\geqslant 1} x^{k-1}/(k-1)! = e^x$,

$$g_\alpha(x,y) \leqslant 2x^{\alpha-1}(e^{-x} + x); \tag{13}$$

substituting into Theorem 3.6,

$$h_\alpha(x) \leqslant (2/\mu)^n \prod_i (x_i/\mu)^{\alpha_i - 1}(1 + x_i/\mu). \tag{14}$$

When $\alpha_i > 1$, the factor $(x_i/\mu)^{\alpha_i-1} \to 0$ as $x_i \to 0+$, yielding $h_\alpha(x) \to 0$.

Derivative formula. Differentiating the series in Theorem 3.4 term by term (using $\Gamma(k+\alpha) = (k+\alpha-1)\Gamma(k+\alpha-1)$, valid when $\alpha > 0$),

$$\partial_x g_\alpha(x, y) = g_{\alpha-1}(x, y) - g_\alpha(x, y), \qquad \alpha > 1, \tag{15}$$

with convergence locally uniform by (13) applied to $g_{\alpha-1}$ (valid since $\alpha - 1 > 0$). Inserting into Theorem 3.6: the integrand and its x_i-derivative are bounded by an integrable function of (x, ω) uniformly in x_i on compact subsets of $(0, \infty)$ (use (13) for $g_{\alpha_i-1}(\cdot, Y_i)$ and $g_{\alpha_i}(\cdot, Y_i)$), justifying dominated convergence to differentiate under E, which gives (10).

(iii). Iterating (10),

$$\frac{\partial^{|J|}}{\partial x_J} h_\alpha = \sum_{\delta \in \{0,1\}^J} (-1)^{|J|-|\delta|} h_{\alpha - \sum_{i \in J} \delta_i e_i}, \tag{16}$$

each summand in L_1 by (i) (all $\alpha_i - \delta_i \geqslant \alpha_i - 1 > 0$). Integration by parts in each x_j, $j \in J$: the boundary term at $x_j \to \infty$ vanishes by (14) (exponential decay in every coordinate via $\int_0^\infty x^{\alpha-1}(1+x)e^{-ax}\,\mathrm{d}x < \infty$, implying pointwise decay of the partial integral); at $x_j \to 0+$ by (ii), applied to the $\alpha'_j := \alpha_j - \delta_j \geqslant \alpha_j - 1 > 0$ shape, together with the extra factor $e^{-\sum_i \lambda_i x_i}$ which is bounded. Each integration by parts brings down a λ_j, giving (11). □

3.5 The family $h_{k,C}$ and its Laplace transform

Definition 3.8. For $C \succ 0$ and $\mu > 0$ with $C - \mu I_n \succ 0$, write $C = \mu I_n + AA^\top$. Let $(g_j^{(l)})$, $j \leqslant n$, $l \leqslant k$, be i.i.d. $\mathcal{N}(0, 1)$, and

$$Y_i := \sum_{l=1}^{k} \Big(\sum_{j=1}^{n} \tfrac{1}{\sqrt{2\mu}} g_j^{(l)} a_{ij} \Big)^2, \qquad h_{k,C} := h_{(k/2,\ldots,k/2),\mu,Y}. \tag{17}$$

Lemma 3.9. For $\lambda_i \geqslant 0$, $\int e^{-\sum_i \lambda_i x_i} h_{k,C}\,\mathrm{d}x = |I_n + \Lambda C|^{-k/2}$.

Proof. Step 1: scalar Laplace transform. Substitute $u := x/\mu$ and expand the series in Theorem 3.4:

$$\begin{aligned}
\int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y)\,\mathrm{d}x &= \int_0^\infty e^{-\lambda\mu u - u - y} \sum_{k \geqslant 0} \tfrac{u^{k+\alpha-1}}{\Gamma(k+\alpha)} \tfrac{y^k}{k!}\,\mathrm{d}u \\
&= e^{-y} \sum_{k \geqslant 0} \tfrac{y^k}{k!} (1 + \mu\lambda)^{-(k+\alpha)} \\
&= (1 + \mu\lambda)^{-\alpha} \exp\big(-\tfrac{\mu\lambda}{1+\mu\lambda} y\big),
\end{aligned}$$

using $\int_0^\infty u^{k+\alpha-1} e^{-(1+\mu\lambda)u}\,\mathrm{d}u = \Gamma(k+\alpha)(1+\mu\lambda)^{-(k+\alpha)}$ (Fubini-term-by-term justified by absolute convergence). So

$$\int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y)\,\mathrm{d}x = (1 + \mu\lambda)^{-\alpha} \exp\big(-\tfrac{\mu\lambda}{1+\mu\lambda} y\big). \tag{18}$$

Step 2: product form. Applying (18) coordinatewise with $\alpha_i = k/2$ and Fubini (nonneg integrand),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C}\,\mathrm{d}x = \prod_i (1 + \mu\lambda_i)^{-k/2} \cdot \mathsf{E} \exp\Big(-\sum_i \tfrac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\Big). \tag{19}$$

Step 3: evaluate the expectation. From (17), set $X^{(l)} := \frac{1}{\sqrt{2\mu}} A g^{(l)}$, so $X^{(l)} \sim \mathcal{N}(0, \frac{1}{2\mu} AA^\top)$ i.i.d. across l and $Y_i = \sum_{l=1}^k (X_i^{(l)})^2$. By independence across l and Theorem 3.1 applied to each $X^{(l)}$ with $\tilde{\Lambda} := \mathrm{diag}(\frac{\mu\lambda_i}{1+\mu\lambda_i})$,

$$\mathsf{E} \exp\Big(-\sum_i \tfrac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\Big) = \Big(\mathsf{E} \exp(-\langle \tilde{\Lambda} X^{(1)}, X^{(1)} \rangle)\Big)^k = |I_n + \tfrac{1}{\mu} \tilde{\Lambda} AA^\top|^{-k/2}.$$

Step 4: combine. Using $(I_n + \mu\Lambda)\tilde{\Lambda} = \mu\Lambda$ (entrywise diagonal check), so $\tilde{\Lambda} = (I_n + \mu\Lambda)^{-1}\mu\Lambda$:

$$\prod_i (1 + \mu\lambda_i)^{-k/2} = |I_n + \mu\Lambda|^{-k/2},$$

and $|I_n + \mu\Lambda| \cdot |I_n + \frac{1}{\mu}\tilde{\Lambda} AA^\top| = |I_n + \mu\Lambda + \Lambda AA^\top| = |I_n + \Lambda(\mu I_n + AA^\top)| = |I_n + \Lambda C|$, giving the claim. □

3.6 Uniqueness of the Laplace transform

Theorem 3.10. If $\varphi_1, \varphi_2 \in L_1((0,\infty)^n)$ satisfy $\int e^{-\sum_i \lambda_i x_i} \varphi_1 \, \mathrm{d}x = \int e^{-\sum_i \lambda_i x_i} \varphi_2 \, \mathrm{d}x$ for all $\lambda \in [0,\infty)^n$, then $\varphi_1 = \varphi_2$ a.e.

Proof. Put $\varphi := \varphi_1 - \varphi_2$ and $y_i := e^{-x_i}$, mapping $(0,\infty)^n \to (0,1]^n$ with Jacobian $\prod_i y_i^{-1}$. The hypothesis becomes $\int_{(0,1]^n} y^\lambda \tilde{\varphi}(y) \, \mathrm{d}y = 0$ for all $\lambda \in [0,\infty)^n$, where $\tilde{\varphi}(y) = \varphi(-\log y) \prod_i y_i^{-1} \in L_1((0,1]^n)$. Polynomials (integer λ) are dense in $C([0,1]^n)$ by Stone–Weierstrass, hence $\int \psi\tilde{\varphi} = 0$ for all continuous ψ, forcing $\tilde{\varphi} = 0$ and $\varphi = 0$ a.e. □

4 Proof of Theorem 2.4

4.1 Reduction to monotonicity

With $s_i := \frac{1}{2} t_i^2$, $K := [0, s_1] \times \cdots \times [0, s_n]$, and $Z(\tau)$ as above, (2) is equivalent to $\mathsf{P}(Z(1) \in K) \geqslant \mathsf{P}(Z(0) \in K)$. At $\tau = 0$, $C(0) = \mathrm{diag}(C_{11}, C_{22})$ is block-diagonal, so the two blocks are independent and the RHS factorizes. At $\tau = 1$ we recover C. Hence Theorem 2.4 follows from:

Proposition 4.1. $\tau \mapsto \int_K f(x,\tau) \, \mathrm{d}x$ is non-decreasing on $[0,1]$.

4.2 Laplace-transform computation of $\partial_\tau f$

Strategy. Theorem 4.1 amounts to $\int_K \partial_\tau f(x,\tau) \, \mathrm{d}x \geqslant 0$ for each $\tau \in (0,1)$ (Theorem 3.3(ii) with $\lambda = 0$ justifies $\partial_\tau \int_K f = \int_K \partial_\tau f$). We obtain this by: (a) computing the Laplace transform of $\partial_\tau f$ and matching it with that of a sum of derivatives of h_τ; (b) using Laplace injectivity (Theorem 3.10) to get an a.e. pointwise identity; (c) integrating this identity over K.

Step (a). Since $Z_i(\tau) = \frac{1}{2} X_i(\tau)^2$, applying Theorem 3.1 to $X(\tau) \sim \mathcal{N}(0, C(\tau))$ with $\lambda_i/2$ in place of λ_i,

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} f(x,\tau) \, \mathrm{d}x = \mathsf{E} e^{-\sum_i \lambda_i Z_i(\tau)} = |I_n + \Lambda C(\tau)|^{-1/2}.$$

Differentiating in τ, with Theorem 3.3(ii) (taking $K = [0,\infty)^n$) justifying the interchange:

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x,\tau) \, \mathrm{d}x = \partial_\tau |I_n + \Lambda C(\tau)|^{-1/2}. \tag{20}$$

Expand via Theorem 3.2(i) applied to $A = \Lambda C(\tau)$:

$$|I_n + \Lambda C(\tau)| = 1 + \sum_{\varnothing \neq J \subseteq [n]} |(\Lambda C(\tau))_J| = 1 + \sum_{\varnothing \neq J \subseteq [n]} |C(\tau)_J| \prod_{j \in J} \lambda_j, \tag{21}$$

using $|(\Lambda C(\tau))_J| = |\Lambda_J||C(\tau)_J| = \prod_{j \in J} \lambda_j \cdot |C(\tau)_J|$ since Λ is diagonal.

The block structure of $|C(\tau)_J|$. For $J \subseteq [n]$, set $J_1 := [n_1] \cap J$ and $J_2 := J \setminus [n_1]$. If $J_1 = \varnothing$ or $J_2 = \varnothing$, then $C(\tau)_J = C_J$ is τ-independent, since the off-diagonal blocks of $C(\tau)$ (where the τ appears) are not sampled. Otherwise, writing $C(\tau)_J$ in block form with blocks indexed by J_1, J_2,

$$C(\tau)_J = \begin{pmatrix} C_{J_1} & \tau C_{J_1 J_2} \\ \tau C_{J_2 J_1} & C_{J_2} \end{pmatrix},$$

and applying Theorem 3.2(ii),

$$|C(\tau)_J| = |C_{J_1}||C_{J_2}| \prod_{i=1}^{|J_1|} (1 - \tau^2 \mu_{J_1,J_2}(i)), \tag{22}$$

where the $\mu_{J_1,J_2}(i) \in [0,1]$ are the eigenvalues of $C_{J_1}^{-1/2} C_{J_1 J_2} C_{J_2}^{-1} C_{J_2 J_1} C_{J_1}^{-1/2}$ (bounded by Theorem 3.2(ii) since $C_J \succ 0$). Differentiating the product in τ,

$$\partial_\tau |C(\tau)_J| = -|C_{J_1}||C_{J_2}| \sum_{i=1}^{|J_1|} 2\tau \mu_{J_1,J_2}(i) \prod_{l \neq i} (1 - \tau^2 \mu_{J_1,J_2}(l)) \leqslant 0 \quad \text{on } [0,1].$$

Define

$$a_J(\tau) := -\partial_\tau |C(\tau)_J| \geqslant 0 \quad (\tau \in [0,1]), \tag{23}$$

with $a_J := 0$ if $J_1 = \varnothing$ or $J_2 = \varnothing$. Then

$$\partial_\tau |I_n + \Lambda C(\tau)| = -\sum_{\varnothing \neq J \subseteq [n]} a_J(\tau) \prod_{j \in J} \lambda_j. \tag{24}$$

Combining the chain rule $\partial_\tau |\cdot|^{-1/2} = -\frac{1}{2} |\cdot|^{-3/2} \partial_\tau |\cdot|$ with (20) and (24):

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x,\tau)\, \mathrm{d}x = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) |I_n + \Lambda C(\tau)|^{-3/2} \prod_{j \in J} \lambda_j. \tag{25}$$

4.3 Identification via $h_\tau := h_{3,C(\tau)}$

By Theorem 3.9 with $k = 3$ and Theorem 3.7(iii) (valid since $\alpha_i = \frac{3}{2} > 1$),

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\tau \, \mathrm{d}x = \Big(\prod_{j \in J} \lambda_j\Big) |I_n + \Lambda C(\tau)|^{-3/2}.$$

Comparing with (25): $\partial_\tau f(\cdot,\tau)$ and $\sum_J \frac{1}{2} a_J(\tau)\, \partial^{|J|} h_\tau / \partial x_J$ are L_1 with the same Laplace transform, so by Theorem 3.10,

$$\partial_\tau f(x,\tau) = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) \tfrac{\partial^{|J|}}{\partial x_J} h_\tau(x) \quad \text{a.e. on } (0,\infty)^n. \tag{26}$$

4.4 Integration over K

Integrate (26) over $K = \prod_j [0, s_j]$. For $J^c := [n] \setminus J$ and $(s_J, x_{J^c})_i = s_i$ $(i \in J)$, $= x_i$ $(i \in J^c)$, the fundamental theorem of calculus in each x_j, $j \in J$, gives

$$\int_K \tfrac{\partial^{|J|}}{\partial x_J} h_\tau(x)\,\mathrm{d}x = \int_{\prod_{j\in J^c}[0,s_j]} h_\tau(s_J, x_{J^c})\,\mathrm{d}x_{J^c} \geqslant 0, \tag{27}$$

since the lower-endpoint contributions vanish by Theorem 3.7(ii) ($\alpha_j = \frac{3}{2} > 1$), and $h_\tau \geqslant 0$. Since also $a_J \geqslant 0$,

$$\int_K \partial_\tau f(x,\tau)\,\mathrm{d}x = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) \int_{\prod_{j\in J^c}[0,s_j]} h_\tau(s_J, x_{J^c})\,\mathrm{d}x_{J^c} \geqslant 0,$$

which proves Theorem 4.1 and hence Theorem 2.4. □

5 Two corollaries

Corollary 5.1 (Šidák)**.** For $X \sim \mathcal{N}(0, C)$ and $s_i \geqslant 0$,

$$\mathsf{P}(|X_i| \leqslant s_i,\ i = 1, \dots, n) \geqslant \prod_{i=1}^{n} \mathsf{P}(|X_i| \leqslant s_i).$$

Proof. Induct on n. Case $n = 1$ is trivial. For the step, take $K_1 := \{|x_1| \leqslant s_1\}$ and $L_1 := \{|x_i| \leqslant s_i, 2 \leqslant i \leqslant n\}$ (both closed symmetric convex). Theorem 2.4 (applied with $n_1 = 1$, $n_2 = n - 1$) gives $\mathsf{P}(X \in K_1 \cap L_1) \geqslant \mathsf{P}(X \in K_1)\mathsf{P}(X \in L_1)$; apply induction to $\mathsf{P}(X \in L_1)$. □

Corollary 5.2 (Decoupling)**.** For $X \sim \mathcal{N}(0, C)$ and closed symmetric convex $K, L \subseteq \mathbb{R}^n$,

$$\mathsf{P}(X \in K \cap L) \geqslant \mathsf{P}(X \in K)\,\mathsf{P}(X \in L).$$

Proof. Restatement of Theorem 1.1 for this K, L. □

6 Notes

6.1 Historical remarks

The conjecture originates in statistical work on joint coverage of rectangular confidence regions for multivariate normal means: Dunnett–Sobel (1955), Dunn (1958). Das Gupta, Eaton, Olkin, Perlman, Savage, and Sobel (1972) stated it in geometric form. Partial results before Royen:

- Khatri (1967) and Šidák (1967), independently: one set is a symmetric strip (= intersection of symmetric slabs). This is our Theorem 5.1.
- Pitt (1977): $d = 2$.
- Schechtman–Schlumprecht–Zinn (1998): both sets are centered ellipsoids.
- Hargé (1999): one set is a centered ellipsoid.

Royen (2014) proved the full conjecture and, more generally, the analogous inequality for multivariate gamma vectors $\Gamma_n(\alpha, R)$ with $\alpha > 0$ half-integer; the Gaussian case is $\alpha = \frac{1}{2}$. Latała–Matlak (2015) specialised and reorganised the argument for the Gaussian case, which is the version followed here.

6.2 Extension to singular covariance

Let $C \succeq 0$ with $\operatorname{rank} C = r \leqslant n$. Spectrally, $C = BB^\top$ where $B \in M_{n\times r}$ has orthonormal columns scaled by $\sqrt{\text{eigenvalues}}$; B has full column rank. If $G \sim \mathcal{N}(0, I_r)$, then $BG \sim \mathcal{N}(0, BB^\top) = \mathcal{N}(0, C)$.

For closed symmetric convex $K \subseteq \mathbb{R}^n$:

- $B^{-1}(K) = \{g \in \mathbb{R}^r : Bg \in K\}$ is closed by continuity of $g \mapsto Bg$;
- convex, since $Bg_1, Bg_2 \in K$ implies $B(\alpha g_1 + (1-\alpha)g_2) = \alpha Bg_1 + (1-\alpha)Bg_2 \in K$ by convexity of K;
- symmetric, since $g \in B^{-1}(K) \iff Bg \in K \iff -Bg \in K \iff -g \in B^{-1}(K)$.

Hence, writing $\mu = \mathcal{N}(0, C)$ and $\nu = \mathcal{N}(0, I_r)$,

$$\mu(K \cap L) = \mathsf{P}(BG \in K \cap L) = \nu(B^{-1}(K) \cap B^{-1}(L)) \geqslant \nu(B^{-1}(K))\nu(B^{-1}(L)) = \mu(K)\mu(L),$$

where the inequality is Theorem 1.1 in dimension r (non-singular case, already established).

6.3 The choice $k = 3$

Two constraints pin down the shape parameter.

Laplace-side matching. Differentiating $|I + \Lambda C(\tau)|^{-1/2}$ in τ yields a linear combination of $|I + \Lambda C(\tau)|^{-3/2} \prod_{j\in J} \lambda_j$ ((25)). To rewrite this as the Laplace transform of a density, we need an auxiliary h_τ whose Laplace transform is $|I + \Lambda C(\tau)|^{-3/2}$, i.e., $h_\tau = h_{k,C(\tau)}$ with $k = 3$ in Theorem 3.9.

Integration by parts. Theorem 3.7(ii)–(iii) require $\alpha_i = k/2 > 1$ both for $h_\alpha(x) \to 0$ as $x_i \to 0+$ and for validity of the derivative identity $\partial_x g_\alpha = g_{\alpha-1} - g_\alpha$. $k = 3$ gives $\alpha_i = \frac{3}{2} > 1$.

The method applies whenever the Laplace transform of the underlying squared vector has the form $|I + \Lambda C|^{-\beta}$ with $\beta + 1 > 1$, i.e., $\beta > 0$; in the Gaussian case $\beta = \frac{1}{2}$. This is the range covered by Royen's multivariate gamma result.

6.4 Strict positivity of a_J

Under $C \succ 0$, the matrix $C_{J_1}^{-1/2} C_{J_1J_2} C_{J_2}^{-1} C_{J_2J_1} C_{J_1}^{-1/2}$ is the matrix of squared canonical correlations between the coordinate blocks J_1 and J_2. Its eigenvalues $\mu_{J_1,J_2}(i)$ satisfy $\mu_{J_1,J_2}(i) \in [0, 1)$ strictly, since $|C(\tau)_J| > 0$ for all $\tau \in [0, 1]$ when $C \succ 0$ (the endpoints $\tau = \pm 1$ would be needed to attain $\mu = 1$). Consequently, for any J with $J_1, J_2 \neq \varnothing$ and $C_{J_1J_2} \neq 0$, some $\mu_{J_1,J_2}(i) > 0$, so $a_J(\tau) > 0$ on $(0, 1]$. At $\tau = 0$, every $a_J(0) = 0$ because each $|C(\tau)_J|$ is even in τ (see (22)).

6.5 Equality in the GCI

Proposition 6.1. Let $X \sim \mathcal{N}(0, C)$, $C \succ 0$, and let $K, L \subseteq \mathbb{R}^n$ be as in Theorem 2.4. Then equality $\mathsf{P}(X \in K \cap L) = \mathsf{P}(X \in K)\mathsf{P}(X \in L)$ holds iff $C_{12} = 0$.

Proof. If $C_{12} = 0$, then $C(\tau) = \operatorname{diag}(C_{11}, C_{22})$ for all τ, so the two coordinate blocks of X are independent and both sides of (2) are equal.

Conversely, suppose equality. From the fundamental theorem of calculus and (26):

$$\mathsf{P}(X \in K \cap L) - \mathsf{P}(X \in K)\mathsf{P}(X \in L) = \int_0^1 \sum_J \tfrac{1}{2} a_J(\tau) \int_{\prod_{J^c}[0,s_j]} h_\tau(s_J, x_{J^c})\, \mathrm{d}x_{J^c}\, \mathrm{d}\tau.$$

Each a_J is polynomial in τ (since $|C(\tau)_J|$ is polynomial in the entries, which are affine in τ), hence continuous on $[0, 1]$ with $a_J \geqslant 0$. The inner integral is > 0 for every $\tau \in [0, 1]$: $g_{3/2}(\cdot, y) > 0$ on

$(0,\infty)$ for each $y \geqslant 0$, because every term in the series of Theorem 3.4 is nonnegative and the $k=0$ term equals $x^{1/2}e^{-x-y}/\Gamma(3/2) > 0$; taking expectation over Y preserves strict positivity, so $h_\tau > 0$ on $(0,\infty)^n$ and the integral over any product $\prod_{J^c}[0,s_j]$ of positive lengths is > 0.

Equality thus forces $a_J(\tau) \equiv 0$ on $[0,1]$ (by continuity) for every J with $J_1, J_2 \neq \varnothing$. By Section 6.4, this means $C_{J_1J_2} = 0$ for every such J. Taking $J = \{i,j\}$ with $i \leqslant n_1 < j \leqslant n$ yields $C_{ij} = 0$, so $C_{12} = 0$. □

6.6 Verification of (27) for $n = 2$

For $J = \{1\}$:

$$\int_0^{s_1}\int_0^{s_2} \partial_{x_1} h_\tau \,\mathrm{d}x_1\,\mathrm{d}x_2 = \int_0^{s_2} \big(h_\tau(s_1,x_2) - h_\tau(0+,x_2)\big)\mathrm{d}x_2 = \int_0^{s_2} h_\tau(s_1,x_2)\,\mathrm{d}x_2.$$

For $J = \{1,2\}$: $\int_0^{s_1}\int_0^{s_2} \partial^2_{x_1x_2} h_\tau = h_\tau(s_1,s_2) - h_\tau(s_1,0+) - h_\tau(0+,s_2) + h_\tau(0+,0+) = h_\tau(s_1,s_2)$. Both use $h_\tau(\cdot) \to 0$ at $0+$ from Theorem 3.7(ii).

6.7 Necessity of symmetry

Without $K = -K$, $L = -L$ the inequality fails. Under $\mu = \mathcal{N}(0, I_2)$ take

$$K := \{x : x_1 \geqslant 1\}, \qquad L := \{x : x_1 \leqslant -1\}.$$

Both are closed convex, neither is symmetric. Then $\mu(K) = \mu(L) = 1 - \Phi(1) \approx 0.1587$, so $\mu(K)\mu(L) \approx 0.0252$, while $K \cap L = \varnothing$ gives $\mu(K \cap L) = 0 < \mu(K)\mu(L)$.

6.8 Šidák correction for multiple testing

Let $Z_1, \ldots, Z_n$ be jointly Gaussian with $Z_i \sim \mathcal{N}(0,1)$ marginally. By Theorem 5.1,

$$\mathsf{P}(|Z_i| > c \text{ for some } i) \leqslant 1 - (2\Phi(c) - 1)^n,$$

independently of the joint covariance. Setting the right side equal to α gives $c = \Phi^{-1}\big(\frac{1+(1-\alpha)^{1/n}}{2}\big)$, the Šidák threshold.

6.9 Related correlation inequalities

	Measure	Events/functions	Conclusion
GCI	Centered Gaussian on $\mathbb{R}^n$	Symmetric convex K, L	$\mu(K \cap L) \geqslant \mu(K)\mu(L)$
FKG (1971)	Log-concave on a lattice	Monotone f, g	$\mathsf{E}[fg] \geqslant \mathsf{E}f \cdot \mathsf{E}g$
Harris (1960)	Product on $\{0,1\}^n$	Monotone events A, B	$\mathsf{P}(A \cap B) \geqslant \mathsf{P}(A)\mathsf{P}(B)$

FKG contains Harris as a special case. The GCI is formally unrelated to FKG: symmetric convex sets in $\mathbb{R}^n$ are not monotone for any coordinate order, and monotone events on a lattice are not symmetric.

6.10 Open questions

Quantitative GCI. No explicit nontrivial lower bound on the excess $\mu(K \cap L) - \mu(K)\mu(L)$ is known in the generality of Theorem 1.1. Candidates depending on Gaussian widths, eccentricity, and the spectral gap of C have been proposed but not established.

Non-Gaussian extensions. Counterexamples rule out general log-concave measures. The exact class of product-type inequalities valid for symmetric convex sets — beyond the Gaussian and multivariate-gamma families of Royen — is open.

Functional forms. Whether $\mathsf{E}[f(X)g(X)] \geqslant \mathsf{E}f(X)\,\mathsf{E}g(X)$ holds for centered Gaussian X and even quasi-concave $f, g \geqslant 0$ is open; partial results are in Latała–Matlak (2015, Remark 3).

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